

Systematic Bitcoin Futures Mean-Reversion Research (2020-2026)

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Executive Summary

This project began as an attempt to develop a statistically robust mean-reversion strategy for Bitcoin futures using Binance 5-minute data. The research evolved from a simple higher-timeframe (HTF) VWAP mean-reversion concept into a comprehensive systematic trading research effort involving custom backtesting infrastructure, walk-forward validation, large-scale parameter optimization, regime analysis, trade expectancy studies, and hypothesis-driven strategy refinement.

Over the course of development, approximately 2,100 lines of Python code were written, more than 25,000 parameter combinations were evaluated, and 324 trades were analyzed across roughly five years of market data. The final strategy produced a Sharpe ratio of 0.72, maximum drawdown of -23.7%, and total return of 132.3%, compared to Bitcoin buy-and-hold returns of 525.3% over the same period.

While the final strategy did not demonstrate sufficient robustness to warrant deployment, the research process generated valuable insights regarding regime dependence, parameter stability, and the challenges of designing durable mean-reversion systems in cryptocurrency markets.

Initial Hypothesis

The original hypothesis was straightforward:

Large deviations from a higher-timeframe VWAP should eventually mean revert.

The first version of the strategy used:

- 60-day VWAP
- Distance-from-VWAP z-scores
- Mean-reversion entries
- Mean-reversion exits

The concept appeared reasonable from a market-structure perspective:

- Extreme dislocations often attract liquidity.
 - Market participants may rebalance toward longer-term fair value.
 - VWAP can act as a proxy for consensus price.
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First Major Finding: Statistical Insignificance

The first strategy produced what initially appeared to be encouraging results:

- Sharpe ratio higher than buy-and-hold
- Significantly lower drawdown
- Positive returns

However, the strategy only generated:

- 10 trades
- Across approximately two years

This immediately created a problem.

Even though the performance metrics appeared attractive, the sample size was far too small to draw meaningful conclusions.

A strategy that earns a Sharpe ratio from 10 trades cannot be considered statistically reliable.

This realization became the first major turning point in the project.

Instead of attempting to optimize the existing strategy further, the decision was made to redesign the framework around a lower-timeframe VWAP approach that would generate significantly more observations.

Transition to Shorter-Term VWAP Research

The strategy was redesigned using:

- Shorter VWAP windows
- Lower z-score thresholds
- Additional trend filters
- Higher trading frequency

The objective was not necessarily to increase returns.

The objective was to increase the number of independent observations available for research.

This transition ultimately increased trade count from roughly:

10 trades

to:

324 trades

across the full test period.

This dramatically improved the ability to:

- perform sensitivity analysis
- run walk-forward tests
- study parameter stability
- investigate failure modes

Research Infrastructure

To support this process, a custom research framework was built.

The system combined:

Vectorized Research Components

Used for:

- Rolling VWAP calculations
- Moving averages
- Standard deviation calculations
- Volume calculations
- Performance metrics
- Equity curve generation

Event-Driven Execution Logic

Used for:

- Position state tracking
- Entry management
- Exit management
- Maximum hold tracking
- Trend filtering
- Trade lifecycle management

This architecture allowed the strategy to behave realistically while maintaining acceptable research speed.

Walk-Forward Validation Framework

A major emphasis of the project was avoiding overfitting.

Instead of selecting parameters and evaluating them on a single dataset, a walk-forward framework was developed.

The process consisted of:

Training Window

10 months

Out-of-Sample Windows

Six consecutive:

10-month periods

This created a sequence of entirely unseen datasets used to evaluate robustness.

Rather than asking:

"What worked best?"

the research focused on:

"What continues to work after optimization?"

Parameter Optimization

A multiprocessing grid-search system was developed using:

- ProcessPoolExecutor
- Per-worker caching
- Fast metric calculations

The system evaluated more than:

25,000+

parameter combinations.

Parameters included:

- VWAP length
- Moving-average length
- Entry z-score
- Exit z-score
- Maximum hold period

Metrics evaluated included:

- Sharpe
 - Calmar
 - Drawdown
 - Trade count
 - Return
 - Holding period statistics
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Early Walk-Forward Success

The first results were encouraging.

The optimized strategies:

- Survived training
- Performed well in the first 10-month out-of-sample window

- Performed well in the second 10-month out-of-sample window

At this stage the research appeared promising.

Many parameter sets continued producing positive risk-adjusted returns on unseen data.

This suggested that the optimization process had identified genuine market behavior rather than pure noise.

Regime Breakdown

The next stage produced the most important finding of the project.

After surviving:

- Training
- OOS Window 1
- OOS Window 2

The strategy family collapsed.

Virtually all previously successful parameter combinations failed simultaneously.

This was especially notable because the failure coincided with the beginning of a major Bitcoin bull market phase.

This suggested that the strategy's edge was not universally persistent.

Instead, it appeared highly dependent on market regime.

The implication was significant:

The strategy was not merely experiencing random variance.

The underlying market dynamics had changed.

This became the central research question for the remainder of the project.

Investigating Failure Modes

Instead of immediately searching for new parameters, the research shifted toward understanding why the strategy was failing.

Several hypotheses were investigated.

Hypothesis 1: Holding Time

Reviewing trade reports revealed a pattern:

Winning trades tended to resolve quickly.

Losing trades tended to remain open significantly longer.

This observation motivated a trade-duration study.

Trade Duration Expectancy Analysis

Trades were grouped into holding-period buckets.

Expectancy was then aggregated across:

- Multiple parameter sets
- Multiple walk-forward windows

The result was remarkably consistent:

Trades held beyond approximately:

500 bars

were net negative.

This finding appeared across numerous strategy variants.

The implication was:

If mean reversion has not occurred within a reasonable timeframe, the trade may no longer be a mean-reversion opportunity.

Instead, it may represent:

- Trend continuation
 - Regime shift
 - Price discovery
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Maximum Hold Constraints

Based on the trade-duration findings, maximum holding-period constraints were introduced.

The expectation was:

- Remove large losing trades
- Improve risk-adjusted returns
- Increase robustness

The results were mixed.

Some windows improved.

Other windows deteriorated.

More importantly:

The strategy became increasingly sensitive to the chosen hold limit.

For example:

- 300-500 bar limits excelled in some periods
- 700-1000 bar limits excelled in others

This raised a concern.

The strategy was becoming more parameter-dependent rather than more robust.

Hypothesis 2: Volume as a Regime Filter

The next hypothesis focused on volume.

The reasoning was:

A genuine breakout into a new trading range is likely to be accompanied by unusually high participation.

If so, high-volume environments may be poor locations for mean-reversion trades.

A volume filter was introduced using:

- Rolling 24-hour quote volume
- Longer-term average volume baselines

The goal was to block entries when the current volume exceeded historical norms.

Volume Filter Results

The volume filters produced some localized improvements.

However, a broader pattern emerged:

As additional filters were added:

- Performance often improved in specific windows.
- Performance often deteriorated in others.
- Parameter stability generally declined.

The strategy increasingly resembled a collection of regime-specific solutions rather than a robust generalized model.

This was an important warning sign.

Final Strategy Results

The final version of the strategy generated:

Performance

- Sharpe: 0.72
- Max Drawdown: -23.72%
- Total Return: 132.25%

Buy & Hold Benchmark

- Sharpe: 0.81
- Max Drawdown: -77.37%
- Total Return: 525.30%

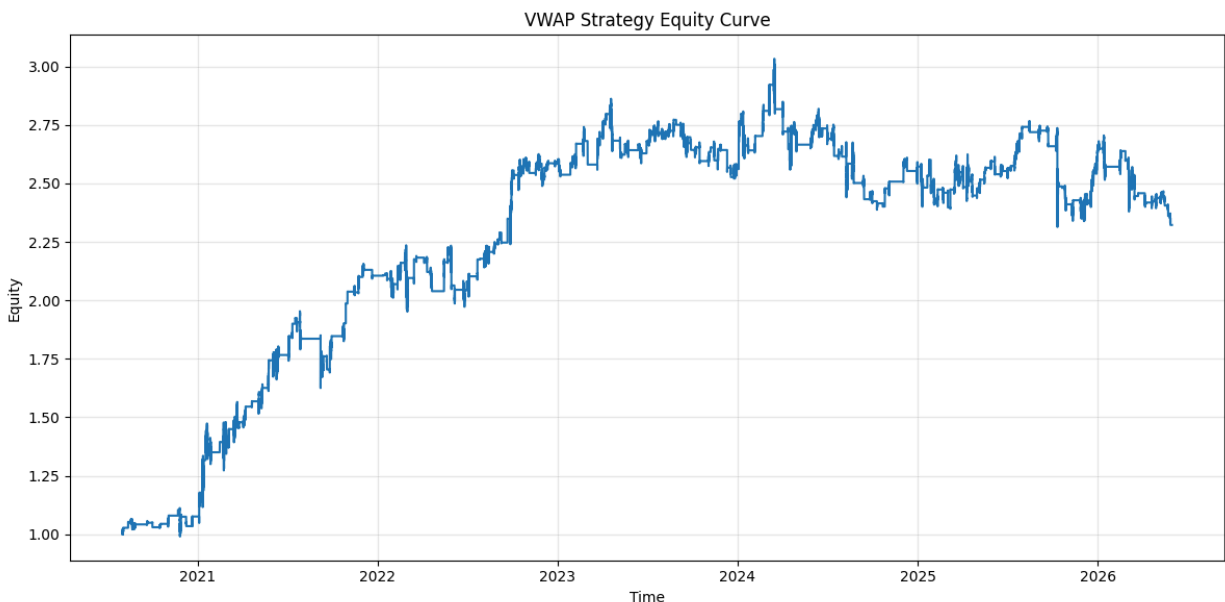
The resulting equity curve demonstrated steady growth through multiple market environments before flattening and becoming increasingly volatile during later regimes, reinforcing the conclusion that the strategy's edge was not stable across all conditions.

The final trade set consisted of:

- 324 completed trades
- 189 long trades
- 135 short trades

with significant variation in profitability across market environments.

Equity curve:



Key Conclusions

1. Sample Size Matters More Than Early Performance

The original higher-timeframe strategy produced attractive metrics but only generated 10 trades. Extending the research revealed that the apparent edge was not sufficiently supported by data.

2. Walk-Forward Testing Reveals What Backtests Hide

Many parameter combinations looked attractive in-sample.

Several even survived multiple out-of-sample windows.

Only long-term walk-forward testing exposed the eventual regime breakdown.

3. Parameter Stability Is More Valuable Than Peak Performance

The highest-performing parameter sets were rarely the most durable.

The strongest candidates tended to be those that performed reasonably well across many windows rather than dominating a single period.

4. Long-Duration Trades Were Consistently Harmful

One of the strongest findings of the project was that extended holding periods generally carried negative expectancy.

This observation appeared repeatedly across parameter sets and regimes.

5. Additional Complexity Did Not Create Robustness

Neither maximum-hold constraints nor volume filters fully solved the underlying issue.

In many cases they increased regime sensitivity rather than reducing it.

6. Market Regime Appears Central to Mean-Reversion Performance

The strategy's largest failures occurred during powerful directional moves in Bitcoin.

This suggests that future research should focus less on optimizing entry thresholds and more on identifying:

- Trend regimes
- Volatility regimes
- Liquidity regimes
- Structural market transitions

before applying mean-reversion logic.

Final Assessment

The project ultimately did not produce a strategy that met the standards required for deployment or live trading. However, it successfully achieved its larger objective:

Developing a repeatable systematic research process.

The project demonstrated the complete research lifecycle:

- hypothesis formation
- backtest development
- walk-forward validation
- parameter optimization
- regime analysis
- failure investigation
- strategy refinement
- conclusion revision

Perhaps the most important outcome was learning when to reject a strategy despite substantial effort invested in it. The research repeatedly favored evidence over attachment to prior hypotheses, ultimately concluding that the observed edge was not sufficiently robust to justify confidence in future performance. That conclusion itself was one of the most valuable findings of the entire project.